

Transform Theory Solved Past Paper

Fourier Transforms · Fourier Integrals · Laplace Transforms

CLASS BS-IV & MSc (Final) Regular Students	COMPOSED BY Nadir Ali Khan BS-IV Mathematics	UNIVERSITY SALU Khairpur Dept. of Mathematics
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Date: 26 May 2025
Total Marks: 50 (Attempt any **Four** questions)
Time: 2 Hours

Q.No. 1 — Solve using Fourier Transform: $\partial^2 u / \partial x^2 = \partial u / \partial t$, $-\infty < x < \infty$, $t > 0$

$$(i) \quad u(x, 0) = f(x) = \begin{cases} u_0, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

Solution

Step 1 — Apply Fourier Transform w.r.t. x . Let $\hat{u}(\omega, t) = \int_{-\infty}^{\infty} u e^{-i\omega x} dx$.

$$\mathcal{F}\{u_{xx}\} = -\omega^2 \hat{u}, \quad \mathcal{F}\{u_t\} = \frac{d\hat{u}}{dt}.$$

The PDE becomes the ODE:

$$\frac{d\hat{u}}{dt} + \omega^2 \hat{u} = 0.$$

Separate variables: $\frac{d\hat{u}}{\hat{u}} = -\omega^2 dt$. Integrate both sides:

$$\ln |\hat{u}| = -\omega^2 t + C \implies \hat{u}(\omega, t) = A(\omega) e^{-\omega^2 t}.$$

Applying $t = 0$: $A(\omega) = \hat{u}(\omega, 0)$, so $\hat{u}(\omega, t) = \hat{u}(\omega, 0) e^{-\omega^2 t}$.

Step 2 — Find $\hat{u}(\omega, 0)$:

$$\hat{u}(\omega, 0) = \int_{-1}^1 u_0 e^{-i\omega x} dx = u_0 \left[\frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^1 = \frac{u_0}{-i\omega} (e^{-i\omega} - e^{i\omega}) = \frac{2u_0 \sin \omega}{\omega}.$$

Step 3 — Solution:

$$\hat{u}(\omega, t) = \frac{2u_0 \sin \omega}{\omega} e^{-\omega^2 t}.$$

By inverse Fourier transform:

$$u(x, t) = \frac{u_0}{\pi} \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} e^{-\omega^2 t + i\omega x} d\omega.$$

$$(ii) \quad u(x, 0) = e^{-|x|}$$

Solution

Step 1 — Find $\hat{u}(\omega, 0)$:

$$\hat{u}(\omega, 0) = \int_{-\infty}^{\infty} e^{-|x|} e^{-i\omega x} dx = \int_{-\infty}^0 e^{x(1-i\omega)} dx + \int_0^{\infty} e^{-x(1+i\omega)} dx = \frac{1}{1-i\omega} + \frac{1}{1+i\omega} = \frac{2}{1+\omega^2}.$$

Step 2 — Solution:

$$\hat{u}(\omega, t) = \frac{2}{1 + \omega^2} e^{-\omega^2 t}.$$

$$u(x, t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{-\omega^2 t}}{1 + \omega^2} e^{i\omega x} d\omega.$$

Q.No. 2 — Fourier Cosine and Sine Integral Representations

Recall: $F_c(\omega) = \int_0^\infty f(x) \cos(\omega x) dx$, $F_s(\omega) = \int_0^\infty f(x) \sin(\omega x) dx$, and

$$f(x) = \frac{2}{\pi} \int_0^\infty F_c(\omega) \cos(\omega x) d\omega = \frac{2}{\pi} \int_0^\infty F_s(\omega) \sin(\omega x) d\omega.$$

(i) $f(x) = xe^{-2x}$, $x > 0$

Solution

Derivation of the formulas (differentiation under the integral sign):

We know $\int_0^\infty e^{-ax} \cos(bx) dx = \frac{a}{a^2 + b^2}$. Differentiate both sides w.r.t. a :

$$-\int_0^\infty x e^{-ax} \cos(bx) dx = \frac{(a^2 + b^2) - 2a^2}{(a^2 + b^2)^2} = \frac{b^2 - a^2}{(a^2 + b^2)^2} \implies \int_0^\infty x e^{-ax} \cos(bx) dx = \frac{a^2 - b^2}{(a^2 + b^2)^2}.$$

Similarly from $\int_0^\infty e^{-ax} \sin(bx) dx = \frac{b}{a^2 + b^2}$, differentiate w.r.t. a :

$$\int_0^\infty x e^{-ax} \sin(bx) dx = \frac{2ab}{(a^2 + b^2)^2}.$$

Now apply with $a = 2$, $b = \omega$:

Cosine transform:

$$F_c(\omega) = \int_0^\infty x e^{-2x} \cos(\omega x) dx = \frac{4 - \omega^2}{(4 + \omega^2)^2}.$$

$$x e^{-2x} = \frac{2}{\pi} \int_0^\infty \frac{(4 - \omega^2) \cos(\omega x)}{(4 + \omega^2)^2} d\omega.$$

Sine transform:

$$F_s(\omega) = \int_0^\infty x e^{-2x} \sin(\omega x) dx = \frac{4\omega}{(4 + \omega^2)^2}.$$

$$x e^{-2x} = \frac{2}{\pi} \int_0^\infty \frac{4\omega \sin(\omega x)}{(4 + \omega^2)^2} d\omega.$$

(ii) $f(x) = e^{-x} \cos x, x > 0$

Solution

Use the product-to-sum identity and the result $\int_0^\infty e^{-ax} \cos(bx) dx = \frac{a}{a^2 + b^2},$

$$\int_0^\infty e^{-ax} \sin(bx) dx = \frac{b}{a^2 + b^2}.$$

Write $\cos x \cos(\omega x) = \frac{1}{2}[\cos((\omega - 1)x) + \cos((\omega + 1)x)]$ and $\cos x \sin(\omega x) = \frac{1}{2}[\sin((\omega + 1)x) + \sin((\omega - 1)x)]$.

Cosine transform:

$$F_c(\omega) = \frac{1}{2} \left[\frac{1}{1 + (\omega - 1)^2} + \frac{1}{1 + (\omega + 1)^2} \right].$$

$$e^{-x} \cos x = \frac{1}{\pi} \int_0^\infty \left[\frac{1}{1 + (\omega - 1)^2} + \frac{1}{1 + (\omega + 1)^2} \right] \cos(\omega x) d\omega.$$

Sine transform:

$$F_s(\omega) = \frac{1}{2} \left[\frac{\omega + 1}{1 + (\omega + 1)^2} + \frac{\omega - 1}{1 + (\omega - 1)^2} \right].$$

$$e^{-x} \cos x = \frac{1}{\pi} \int_0^\infty \left[\frac{\omega + 1}{1 + (\omega + 1)^2} + \frac{\omega - 1}{1 + (\omega - 1)^2} \right] \sin(\omega x) d\omega.$$

Q.No. 3 — Solve $u_{xx} = u_t$ ($x > 0, t > 0$) using Laplace Transform

(i) $u(0, t) = u_0, \quad \lim_{x \rightarrow \infty} u(x, t) = u_1, \quad u(x, 0) = u_1$

Solution

Step 1 — Take Laplace w.r.t. t . Let $\bar{U}(x, s) = \mathcal{L}\{u(x, t)\}$:

$$\bar{U}_{xx} - s\bar{U} = -u(x, 0) = -u_1 \implies \bar{U}_{xx} - s\bar{U} = -u_1.$$

Step 2 — Solve the ODE in x :

Characteristic equation of the homogeneous part $\bar{U}_{xx} - s\bar{U} = 0$:

$$r^2 - s = 0 \implies r = \pm\sqrt{s}.$$

Homogeneous solution: $\bar{U}_h = Ae^{\sqrt{s}x} + Be^{-\sqrt{s}x}$.

Particular solution (constant trial $\bar{U}_p = k$): $0 - sk = -u_1 \Rightarrow k = u_1/s$. So $\bar{U}_p = u_1/s$.

General: $\bar{U} = Ae^{\sqrt{s}x} + Be^{-\sqrt{s}x} + u_1/s$.

Step 3 — Apply boundary conditions:

$\lim_{x \rightarrow \infty} \bar{U}$ must be finite $\Rightarrow A = 0$.

At $x = 0$: $\bar{U}(0, s) = \mathcal{L}\{u_0\} = u_0/s \Rightarrow B + u_1/s = u_0/s \Rightarrow B = (u_0 - u_1)/s$.

$$\bar{U}(x, s) = \frac{u_0 - u_1}{s} e^{-\sqrt{s}x} + \frac{u_1}{s}.$$

Step 4 — Inverse Laplace using $\mathcal{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s}\right\} = \text{erfc}\left(\frac{a}{2\sqrt{t}}\right)$:

$$u(x, t) = (u_0 - u_1) \text{erfc}\left(\frac{x}{2\sqrt{t}}\right) + u_1.$$

Check: $u(x, 0) = (u_0 - u_1) \cdot \text{erfc}(\infty) + u_1 = u_1$. ✓ $u(0, t) = (u_0 - u_1) \cdot 1 + u_1 = u_0$. ✓

(ii) $u(0, t) = u_0, \quad \lim_{x \rightarrow \infty} \frac{u(x, t)}{x} = u_1, \quad u(x, 0) = u_1x$

Solution

Step 1 — Take Laplace:

$$\bar{U}_{xx} - s\bar{U} = -u(x, 0) = -u_1x.$$

Step 2 — Solve the ODE:

Characteristic equation: $r^2 - s = 0 \Rightarrow r = \pm\sqrt{s}$, so $\bar{U}_h = Ae^{\sqrt{s}x} + Be^{-\sqrt{s}x}$.

Particular solution (try $\bar{U}_p = ax + b$): $-s(ax + b) = -u_1x \Rightarrow a = u_1/s, b = 0$. So $\bar{U}_p = u_1x/s$.

General: $\bar{U} = Ae^{\sqrt{s}x} + Be^{-\sqrt{s}x} + u_1x/s$.

Step 3 — Apply BCs:

$\lim_{x \rightarrow \infty} \bar{U}/x = u_1/s$ must be finite $\Rightarrow A = 0$.

At $x = 0$: $\bar{U}(0, s) = u_0/s \Rightarrow B = u_0/s$.

$$\bar{U}(x, s) = \frac{u_0}{s} e^{-\sqrt{s}x} + \frac{u_1x}{s}.$$

Step 4 — Inverse Laplace:

$$u(x, t) = u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{t}}\right) + u_1x.$$

Check: $u(x, 0) = u_0 \cdot 0 + u_1x = u_1x$. ✓ $u(0, t) = u_0 \cdot 1 = u_0$. ✓

Q.No. 4 — Inverse Laplace Transform and IVP

(a) Find $\mathcal{L}^{-1}\left\{\frac{s^2 + 1}{s(s-1)(s+1)(s-2)}\right\}$

Solution

Step 1 — Partial fractions:

$$\frac{s^2 + 1}{s(s-1)(s+1)(s-2)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+1} + \frac{D}{s-2}.$$

Cover-up method:

$$A = \frac{s^2 + 1}{(s-1)(s+1)(s-2)} \Big|_{s=0} = \frac{1}{(-1)(1)(-2)} = \frac{1}{2},$$

$$B = \frac{s^2 + 1}{s(s+1)(s-2)} \Big|_{s=1} = \frac{2}{(1)(2)(-1)} = -1,$$

$$C = \frac{s^2 + 1}{s(s-1)(s-2)} \Big|_{s=-1} = \frac{2}{(-1)(-2)(-3)} = -\frac{1}{3},$$

$$D = \frac{s^2 + 1}{s(s-1)(s+1)} \Big|_{s=2} = \frac{5}{(2)(1)(3)} = \frac{5}{6}.$$

Step 2 — Inverse Laplace:

$$\mathcal{L}^{-1}\left\{\frac{1/2}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \mathcal{L}^{-1}\left\{\frac{1/3}{s+1}\right\} + \mathcal{L}^{-1}\left\{\frac{5/6}{s-2}\right\}.$$

$$\mathcal{L}^{-1}\left\{\frac{s^2+1}{s(s-1)(s+1)(s-2)}\right\} = \frac{1}{2} - e^t - \frac{1}{3}e^{-t} + \frac{5}{6}e^{2t}.$$

(b) Solve $y'' + y = \sqrt{2}\sin(\sqrt{2}t)$, $y(0) = 10$, $y'(0) = 0$

Solution

Step 1 — Take Laplace:

$$[s^2Y - 10s - 0] + Y = \sqrt{2} \cdot \frac{\sqrt{2}}{s^2 + 2} = \frac{2}{s^2 + 2}.$$

$$(s^2 + 1)Y = 10s + \frac{2}{s^2 + 2} \implies Y = \frac{10s}{s^2 + 1} + \frac{2}{(s^2 + 1)(s^2 + 2)}.$$

Step 2 — Partial fractions for $\frac{2}{(s^2 + 1)(s^2 + 2)}$. Let $u = s^2$:

$$\frac{2}{(u+1)(u+2)} = \frac{A}{u+1} + \frac{B}{u+2} \implies A = 2, B = -2.$$

$$Y = \frac{10s}{s^2 + 1} + \frac{2}{s^2 + 1} - \frac{2}{s^2 + 2}.$$

Step 3 — Inverse Laplace (using $\mathcal{L}^{-1}\{1/(s^2 + a^2)\} = \sin(at)/a$):

$$y(t) = 10 \cos t + 2 \sin t - \sqrt{2} \sin(\sqrt{2}t).$$

Check: $y(0) = 10 \cdot 1 + 0 - 0 = 10$. ✓ $y'(0) = 0 + 2 - \sqrt{2} \cdot \sqrt{2} = 2 - 2 = 0$. ✓

Q.No. 5 — Evaluate using Laplace Transform (do not integrate first)

(a) $\mathcal{L}\{e^{-t} * e^t \cos t\}$ (convolution)

Solution

By the **Convolution Theorem**: $\mathcal{L}\{f * g\} = F(s) \cdot G(s)$.

Let $f(t) = e^{-t}$ and $g(t) = e^t \cos t$:

$$F(s) = \mathcal{L}\{e^{-t}\} = \frac{1}{s+1}, \quad G(s) = \mathcal{L}\{e^t \cos t\} = \frac{s-1}{(s-1)^2 + 1} = \frac{s-1}{s^2 - 2s + 2}.$$

$$\mathcal{L}\{e^{-t} * e^t \cos t\} = \frac{1}{s+1} \cdot \frac{s-1}{s^2-2s+2} = \frac{s-1}{(s+1)(s^2-2s+2)}.$$

(b) $\mathcal{L}\left\{\int_0^t \sin \tau \cos(t-\tau) d\tau\right\}$

Solution

The integral $\int_0^t \sin \tau \cos(t-\tau) d\tau$ is the **convolution** of $\sin t$ and $\cos t$.

By the Convolution Theorem:

$$\mathcal{L}\{\sin t * \cos t\} = \mathcal{L}\{\sin t\} \cdot \mathcal{L}\{\cos t\} = \frac{1}{s^2+1} \cdot \frac{s}{s^2+1}.$$

$$\mathcal{L}\left\{\int_0^t \sin \tau \cos(t-\tau) d\tau\right\} = \frac{s}{(s^2+1)^2}.$$